Direct quadratic block coding for Con(ZFC)

A fixed equation in ten unrestricted integers of exact degree 237 520 456

Research report and reproducible construction
Prepared for Vladimir Reshetnikov and the ProveIt project
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Unknowns	10, all in $\mathbb{Z}$
Equation	One ordinary integer polynomial
Exact total degree	237 520 456
Binary arithmetic operations	$809 = 222 + 25 + 562$
Source	22 quadratics in 31 positive integers
New coding positions	32 pair-sum-distinct integers in $[0, 799]$

Abstract

We replace the exponentially spaced scalar source coding in ProveIt's ten-integer consistency equation by a direct encoding of its entire quadratic residual system. A finite Sidon set distinguishes all quadratic monomials. Reversed residual blocks share one carry test, while their marked coefficients remain isolated even though the product blocks overlap. This avoids both the quartic sum of squares and the powers-of-five position scheme.

For the same fixed 121×8801 compiler coefficient tuple as the predecessor, the resulting polynomial has ten unrestricted integer unknowns, exact degree 237 520 456, and an 809-operation division-free circuit. The predecessor has degree 589 042 901 992 797 851 562 527 064 and 1 442 operations. The degree ratio is approximately 2.48×10^{18}.

We prove the new source adapter, give the complete final polynomial by finite arithmetic definitions, and compose it with the inherited compiler endpoint and Sun's Lucas/interpolation theorems. The conclusion is ZFC $\vdash [\forall \mathbf{v} \in \mathbb{Z}^{10},\ F_{10}(\mathbf{v}) \neq 0] \leftrightarrow \mathrm{Con(ZFC)}$. The archive includes coefficient bytes, deterministic generators, independent algebraic checks, and exact-degree certificates. This is an ordinary mathematical proof, not a new end-to-end proof-assistant certificate or a claim of global minimality.

Audited against ProveIt commit
`22e75dd6b87b1b199bb08c778d7cf075ae5aae08`.

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1 Result, comparison, and the meaning of simplicity

Theorem 1.1 (Main result)**.** *The definitions in Sections 4 and 5, with the fixed data in Appendix A, define a particular polynomial*

$$F_{10}(f,g,h,k,l,w,x,y,z,m) \in \mathbb{Z}[f,g,h,k,l,w,x,y,z,m]$$

of exact total degree 237 520 456*. Its delivered circuit has* 809 *binary arithmetic operations. For the repository's specified proof predicate, ordinary mathematics formalizable in ZFC proves*

$$\exists (f,g,h,k,l,w,x,y,z,m) \in \mathbb{Z}^{10}\ F_{10} = 0 \quad \Longleftrightarrow \quad \neg\,\mathrm{Con(ZFC)}. \tag{1}$$

Consequently ZFC proves that F_{10} *has no integer zero if and only if* Con(ZFC)*.*

All coefficients are fixed independently of the truth of consistency. There is no unspecified recursively enumerable index, variable exponent, radical, division, or hidden existential witness in the polynomial. The intermediate names are polynomial abbreviations. They are not additional unknowns.

1.1 The comparison that is actually established

The baseline is the structural-weight, merged-square, quintic ten-integer equation in package 31 of the inspected collection [2]. We keep its whole coefficient tuple, its parent positive system, and its final Lucas/norm construction. This isolates the effect of the new source compiler. The

Measure	Package 31	This construction
Unrestricted integer unknowns	10	10
Total degree	$126496 \cdot 5^{31} + 27064$	237 520 456
Binary arithmetic operations	1,442	809
Fixed circuit leaves (nonliteral)	79	17
Weighted source records	402	136
Exponent defining source-base constant	4,269,882	2,134,940
Largest inherited compiler-port bit length	1,070,004	1,070,004

Table 1: A comparison using the same four native compiler ports. Constant-only recipes and reuse are free in both operation counts.

exact old degree is

$$589\,042\,901\,992\,797\,851\,562\,527\,064.$$

Its ratio to the new degree is approximately $2.4799670391 \times 10^{18}$. The circuit saves 633 operations, a 43.8974% reduction. The 17 fixed leaves consist of 16 coefficient recipes and one explicit source-base constant.

These measures are not interchangeable. The collection also contains positive-variable systems with far fewer arithmetic operations, including the retained 16-positive-variable, 7-equation, degree-24, 59-operation system, and the 31-positive-variable quadratic system with 60 operations [1]. Our equation does not improve those operation counts or their maximum equation degrees. Its advance is on the ten-*unrestricted*-integer branch. We do not establish that ten is minimal, that 809 is an optimal circuit size, or that the displayed Sidon set is optimal.

1.2 What is new and what is inherited

The new contribution is the direct quadratic block adapter:

1. retain the 22 residual equations, without a sum of squares;
2. use pair-sum-distinct positions to distinguish their degree-two monomials;
3. test all 22 selected coefficients in one binary no-carry condition;
4. use block stride $2M + 1$, even though complete product blocks overlap;
5. prove the size and positivity hypotheses for arbitrary bounded g, before decoding g.

Sidon sets and the Bose–Chowla construction are classical [5, 6]. The binary-carry and binomial framework comes from Sun [3]. The ten-integer Lucas/interpolation bridge comes from Sun's 2026 preprint [4]. The quintic merger, coprime square merger, separating weights, and compact norm formula are inherited from package 31. The machine/compiler equivalence is inherited from ProveIt. We prove a new adapter to those results, rather than assuming that Sun's original positional lemma automatically permits different positions.

2 Fixed source and dependency boundary

2.1 The actual source proposition

Let $R_1, \dots, R_{22}$ be the quadratic residuals in the 31 positive variables listed in Appendix A. The four fixed ports are

$$(\beta_{\text{native}}, K_0, M_C, M_F) = (\texttt{beta}, \texttt{K}, \texttt{MC}, \texttt{MF_source}).$$

The high mask M_F is already shifted. All four ports come from the same 121×8801, 943-tile compiler instance. Replacing only some ports, or using the unshifted high mask, would specify a different, unjustified instance.

The inherited positive-system/compiler theorem is

$$\begin{aligned} \exists \mathbf{v} \in \mathbb{Z}_{>0}^{31} \bigwedge_{j=1}^{22} R_j(\mathbf{v}) = 0 \quad &\iff \quad \text{K215 halts on its blank tape} \\ &\iff \quad \neg\,\mathrm{Con(ZFC)}. \end{aligned} \tag{2}$$

The tuplewise 32-to-31 reduction, ordinary Pell/compiler proofs, coefficient adapter, and endpoint reference are retained in `vendor/`. In the upstream Lean source, the endpoint theorem is `KEndpoint215.blank_halts_iff_zfc_refutation`; its conclusion uses `BProv ZFCax_s [] Form.fBot`. We have not rebuilt the full Lean development in this delivery. Its inclusion records the exact endpoint and does not turn the new polynomial reduction into a kernel-checked theorem.

Shift every positive variable:

$$\mathbf{v} = \mathbf{n} + \mathbf{1}, \qquad \mathbf{n} \in \mathbb{N}^{31}, \qquad \mathbb{N} = \{0, 1, 2, \dots\}.$$

Reorder the shifted residuals by putting original residual 5 last:

$$\begin{aligned} r_0, \dots, r_{21} = R_1(\mathbf{n}+\mathbf{1}), \dots, R_4(\mathbf{n}+\mathbf{1}), R_6(\mathbf{n}+\mathbf{1}), \dots, \\ R_{22}(\mathbf{n}+\mathbf{1}), R_5(\mathbf{n}+\mathbf{1}). \end{aligned}$$

This changes no common zero. Original residual 5 is $E - X(a - E)$, whose value at the all-one tuple is 1. Thus

$$r_{21}(\mathbf{0}) = 1, \qquad \mathbf{0} \text{ is not a common zero.} \tag{3}$$

The positive constant will also ensure positivity before digit decoding.

2.2 Literal coefficients and a safe bound

Write

$$r_j(\mathbf{n}) = \sum_{|\alpha|\le 2} a_{j,\alpha}\mathbf{n}^{\alpha}, \qquad c_{j,\alpha} = (2-|\alpha|)!\prod_{i=1}^{31}\alpha_i!\; a_{j,\alpha}.$$

Zero coefficients may be omitted. The generated file `generated/source_records.json` lists all 136 nonzero weighted records, including their exact finite recipes in the four ports. An independent verifier reconstructs them from the pinned parent sparse polynomial, with no call to the builder. Set

$$A_{\max} = \max_{j,\alpha}|c_{j,\alpha}|, \qquad \beta_S = 2^{2\,134\,940}.$$

Exact integer evaluation of the delivered coefficient bytes gives

$$\begin{gathered}\operatorname{bitlength}(A_{\max}) = 2\,134\,926, \qquad A_{\max} < 2^{2\,134\,927},\\ \beta_S > 8192A_{\max}.\end{gathered} \tag{4}$$

The final exponent is even, so β_S is a power of 4. This elementary, finite bound is checked as full integer arithmetic, not inferred from modular samples. The large ports are compressed for transport only: the polynomial's coefficients are the specified uncompressed integers.

3 A compact set of quadratic positions

Use the following ordered set $0 = w_0 < w_1 < \cdots < w_{31} = 799$:

$$\begin{aligned}&(0, 10, 35, 55, 62, 105, 144, 145, 159, 168, 232, 245, 306, 323, 325, 353,\\ &\ 371, 374, 382, 494, 499, 525, 566, 578, 610, 647, 681, 685, 741, 777, 783, 799).\end{aligned} \tag{5}$$

Every unordered pair sum $w_i + w_k$, allowing $i = k$, is distinct. There are exactly $\binom{33}{2} = 528$ such sums. In fact they are distinct modulo 1023, a stronger property than needed here.

The certificate is finite and transparent: enumerate all pairs and check that the 528 resulting residues are distinct. Consequently the main theorem does not depend on accepting an unimplemented finite-field search or an optimality claim.

For provenance, the list was obtained from a Bose–Chowla construction over $\mathbf{F}_{2^{10}}$, represented by the polynomial $x^{10} + x^3 + 1$ and primitive element $\theta = x$ (encoded by the binary integer 2), with the subfield of size 32. A cyclic translate gives a diameter-901 list. Multiplication by 47, followed by translation by -342, modulo 1023 gives (5). The independent Sidon verifier reconstructs that provenance and checks the full primitive cycle. Only the final pair certificate is needed for soundness.

The zero position represents the constant 1. It must be included: checking only pair sums of the 31 positive positions would not exclude collisions between constants, linear monomials, and quadratics. For $|\alpha| \le 2$, write

$$e(\alpha) = \sum_{i=1}^{31}\alpha_i w_i.$$

Padding with $2 - |\alpha|$ copies of the zero position shows that $\alpha \mapsto e(\alpha)$ is injective on monomials of degree at most two.

4 The new simultaneous source adapter

4.1 Complete polynomial definitions

Fix

$$M = 799, \quad s = 2M + 1 = 1599, \quad e_j = 2M + js \quad (0 \le j < 22),$$

and

$$h_D = e_{21} = 35177, \quad Q = h_D + 2M = 36775, \quad t_N = M + Q + 2 = 37576.$$

Here h_D is a fixed integer, unrelated to the eventual unknown h. For the two integer unknowns f, g, define, in the displayed order,

$$b = 1 + 3f, \qquad \mathcal{B} = \beta_S b^2, \qquad \ell = \sum_{i=1}^{31} \mathcal{B}^{w_i}, \tag{6}$$

$$D_j(\mathcal{B}) = \sum_{|\alpha|\le 2} c_{j,\alpha} \mathcal{B}^{2M-e(\alpha)}, \qquad \mathcal{D} = \sum_{j=0}^{21} \mathcal{B}^{js} D_j(\mathcal{B}), \tag{7}$$

$$N_0 = \mathcal{B}^{M+1}, \qquad N_1 = 4\mathcal{B}^{Q+1}, \qquad N = N_0 N_1 = 4\mathcal{B}^{t_N}, \tag{8}$$

$$\mathcal{M} = N_0 - 1 - (b-1)\ell, \qquad U_{\mathrm{mask}} = (\mathcal{B}-2)\sum_{j=0}^{21} \mathcal{B}^{e_j}, \qquad \mathcal{T} = \mathcal{M} + N_0 U_{\mathrm{mask}}, \tag{9}$$

$$\mathcal{J} = 2(1+g)^2\mathcal{D} + \sum_{i=0}^{Q} \mathcal{B}^{i+1}, \qquad \mathcal{S} = g + N_0\mathcal{J}, \tag{10}$$

$$\mathcal{R} = (\mathcal{S} + \mathcal{T} + 1)N + \mathcal{T} + 1, \qquad X = (N-1)\mathcal{R}, \qquad Y = N^2, \tag{11}$$

$$\mathscr{C} = b\mathcal{B}^M. \tag{12}$$

Each exponent is a fixed nonnegative integer. All sums are finite and fixed. In particular these formulas are ordinary polynomials. The expression U_{mask} is distinct from the bridge polynomial U in Section 5.

4.2 Why each target coefficient is exactly one residual

Lemma 4.1 (Simultaneous coefficient extraction)**.** *Let*

$$C_{\mathbf{n}}(T) = \left(1 + \sum_{i=1}^{31} n_i T^{w_i}\right)^2, \qquad D(T) = \sum_{j=0}^{21} T^{js} \sum_{|\alpha|\le 2} c_{j,\alpha} T^{2M-e(\alpha)}.$$

For every j, the coefficient of T^{e_j} in $C_{\mathbf{n}}(T)D(T)$ is $2r_j(\mathbf{n})$.

Proof. The coefficient of $T^{e(\alpha)}$ in $C_{\mathbf{n}}$ is

$$\frac{2!}{(2-|\alpha|)!\prod_i \alpha_i!}\,\mathbf{n}^\alpha.$$

This uses the pair-sum property including the zero position. A term from block k can contribute to target e_j only if

$$e(\gamma) - e(\alpha) = (j-k)s.$$

The left side lies in $[-2M, 2M]$, while $s = 2M+1$. Therefore $k = j$. Within that block, injectivity gives $\gamma = \alpha$. Multiplying the two coefficients cancels the factorials and gives $2a_{j,\alpha}\mathbf{n}^\alpha$. Summing proves the claim. □

This is where two separate savings occur. Quadratic monomials require only a B_2 set, whereas squaring the residuals first would require degree-four separation. Further, full product blocks need not be disjoint. The following intervals explain the shorter stride:

block j	reversed source support	product support
j	$[js, js+2M]$	$[js, js+4M]$

The marked target is $js + 2M$. The preceding product block ends at that target minus 1; the next product block starts at that target plus 1. Overlap elsewhere is harmless.

4.3 Bounds before decoding: avoiding a circular proof

Lemma 4.2 (Global source bounds)**.** *Suppose $f \neq 0$, $b = 1 + 3f$ is an integer square, and $0 \le g < 2\mathscr{C}$. Then $f > 0$, $b \ge 4$, and*

$$0 \le g < N_0, \quad \mathcal{D} > 0, \quad 0 < \mathcal{J} < 2\mathcal{B}^{Q+1} < N_1, \quad 0 \le \mathcal{S}, \mathcal{T} < N.$$

Moreover $3 \mid X$, $X > 3b$, and $Y > \max\{b, 256\}$. These statements do not assume that b is a power of two or that g has the intended digit representation.

Proof. A square congruent to 1 modulo 3 is positive; excluding $f = 0$ excludes $b = 1$, so $b \ge 4$. Also $\mathcal{B} > 2b$, hence $g < 2b\mathcal{B}^M < \mathcal{B}^{M+1} = N_0$.

The reversed source blocks in $\mathcal{D}$ occupy disjoint intervals of length $2M + 1$. Each coefficient has absolute value at most $A_{\max}$. By (3), the leading coefficient, at exponent h_D, is exactly 2. Since $\mathcal{B} - 1 > A_{\max}$,

$$\mathcal{D} \ge 2\mathcal{B}^{h_D} - A_{\max}\frac{\mathcal{B}^{h_D} - 1}{\mathcal{B} - 1} > \mathcal{B}^{h_D} > 0.$$

The upper bound

$$\mathcal{D} \le A_{\max}\sum_{i=0}^{h_D}\mathcal{B}^i < \frac{A_{\max}\mathcal{B}^{h_D+1}}{\mathcal{B} - 1}$$

also holds. Since g is integral, $1 + g \le 2b\mathcal{B}^M$. Consequently

$$0 < \mathcal{J} < \frac{8A_{\max}b^2 + \mathcal{B}}{\mathcal{B} - 1}\mathcal{B}^{Q+1} < 2\mathcal{B}^{Q+1}.$$

The final inequality follows from $\mathcal{B} = \beta_S b^2$ and $\beta_S > 8192A_{\max}$.

The base-$\mathcal{B}$ digits of $\mathcal{M}$ are $\mathcal{B} - b$ at source positions and $\mathcal{B} - 1$ at all other positions through M. Thus $0 \le \mathcal{M} < N_0$. Also

$$0 \le U_{\text{mask}} < \mathcal{B}^{Q+1} < N_1.$$

The asserted bounds on $\mathcal{S}, \mathcal{T}$ follow by concatenation. In particular $\mathcal{R} \ge N + 1 > \mathcal{B} \ge b$; the fixed large β_S makes $X > 3b$ and $Y > 256$ immediate.

Finally $b \equiv 1 \pmod 3$, $\beta_S \equiv 1 \pmod 3$, and $N = 4\mathcal{B}^{t_N} \equiv 1 \pmod 3$. Thus $3 \mid N - 1$, so $3 \mid X$. □

The positive final block is useful for a specific reason: it makes $\mathcal{D} > 0$ for every bounded candidate g. Positivity inferred only after decoding would not justify the hypotheses used to obtain that decoding.

4.4 Low digits and all residuals in one carry condition

For nonnegative integers u, v, write $u \perp_2 v$ when their binary addition has no carry; equivalently, their sets of nonzero binary positions are disjoint.

Lemma 4.3 (Decoding the combined mask)**.** *Assume the hypotheses of Lemma 4.2, and assume additionally that b is a power of two. Then*

$$\mathcal{S} \perp_2 \mathcal{T} \iff \begin{cases} g = \displaystyle\sum_{i=1}^{31} n_i\mathcal{B}^{w_i} \text{ for some } 0 \le n_i < b, \\ r_j(\mathbf{n}) = 0 \quad (0 \le j < 22). \end{cases}$$

Proof. Now $\mathcal{B}, N_0, N_1, N$ are powers of two. The global bounds make the low and high parts genuine, nonoverlapping binary blocks, so

$$\mathcal{S} \perp_2 \mathcal{T} \iff (g \perp_2 \mathcal{M}) \text{ and } (\mathcal{J} \perp_2 U_{\text{mask}}).$$

At a source position the low mask digit is $\mathcal{B}-b$, with all the binary positions above the lowest $\log_2 b$ positions occupied. At every other base position its digit is $\mathcal{B}-1$. Thus $g \perp_2 \mathcal{M}$ gives exactly the displayed representation with $0 \le n_i < b$. In particular the constant position zero is forbidden.

On this decoded branch, let θ_i be the coefficients of $C_{\mathbf{n}}(T)D(T)$. The sum of the nonnegative coefficients of $C_{\mathbf{n}}$ is

$$\left(1+\sum_i n_i\right)^2 < (31b)^2.$$

Because the coefficients of D have absolute value at most $A_{\max}$, convolution gives

$$|\theta_i| < 961 A_{\max} b^2 < \mathcal{B}/2.$$

Therefore

$$\mathcal{J}/2 = \sum_{i=0}^{Q} (\mathcal{B}/2+\theta_i)\mathcal{B}^i$$

has genuine base-$\mathcal{B}$ digits strictly between 0 and $\mathcal{B}$. Division by two occurs only in this proof; the defining polynomial $\mathcal{J}$ is integral and division-free.

Doubling both summands preserves the absence of binary carries. Thus the high condition is equivalent to

$$\mathcal{J}/2 \perp_2 (\mathcal{B}/2-1)\sum_{j=0}^{21} \mathcal{B}^{e_j}.$$

At each marked position the mask occupies every binary position below the top bit. A digit in $\{1,\dots,\mathcal{B}-1\}$ avoids this mask if and only if it is exactly $\mathcal{B}/2$. Hence $\theta_{e_j}=0$ for every j, and Lemma 4.1 identifies this as $2r_j(\mathbf{n})=0$. □

4.5 The binomial interface

Lemma 4.4 (Binary binomial test)**.** *If $N=2^u$ and $0 \le S,T < N$, put*

$$X=(N-1)((S+T+1)N+T+1), \qquad Y=N^2.$$

Then

$$Y \mid \binom{2X}{X} \quad\Longleftrightarrow\quad S \perp_2 T.$$

No hypothesis $S+T<N$ is necessary.

Proof. Write $s_2(v)$ for the number of nonzero binary digits of v. The exact identity

$$X=(S+T)N^2+(N-1-S)N+(N-1-T)$$

separates the lower two blocks, even when $S+T \ge N$. Hence

$$s_2(X)=2u+s_2(S+T)-s_2(S)-s_2(T).$$

The factorial valuation identity gives $v_2\binom{2X}{X} = s_2(X)$. Binary addition gives $s_2(S+T) \le s_2(S)+s_2(T)$, with equality exactly when there is no carry. Divisibility by 2^{2u} is therefore equivalent to that equality. □

This is the binary form of the carry mechanism used by Sun [3, Lemmas 2.2 and 2.4]; the displayed proof specifies the exact normalization needed here.

Proposition 4.5 (Source interface)**.** *The polynomials (6)–(12) have the following properties.*

1. *If the r_j have a common zero in $\mathbb{N}^{31}$, then for arbitrarily large positive integers f there is an integer g with*

$$b \le g < \mathscr{C}, \qquad b \text{ a square and a power of } 2, \qquad Y \mid \binom{2X}{X}.$$

2. *Conversely, if $f \neq 0$, $0 \le g < 2\mathscr{C}$, b is a square and a power of 2, and $Y \mid \binom{2X}{X}$, then the r_j have a common zero in $\mathbb{N}^{31}$.*

3. *Whenever $f \neq 0$, b is a square, and $0 \le g < 2\mathscr{C}$, the size bounds and divisibility $3 \mid X$ in Lemma 4.2 hold. The cap has the fixed form*

$$\mathscr{C} = 2^{2\,134\,940M} b^{2M+1}.$$

Proof. For completeness, choose arbitrarily large $b = 4^a$ greater than all n_i, set $f = (b-1)/3$, and encode $g = \sum_i n_i \mathcal{B}^{w_i}$. Equation (3) excludes the zero tuple. All allowed positions are positive, so $g \ge \mathcal{B} \ge b$. Also

$$g \le (b-1)\sum_{i=1}^{31} \mathcal{B}^{w_i} \le (b-1)\sum_{i=0}^{M} \mathcal{B}^i < b\mathcal{B}^M = \mathscr{C};$$

the last inequality follows from $\mathcal{B} > b$. Apply Lemmas 4.3 and 4.4. The reverse implication is those same lemmas in reverse order, using the global bounds first. The last assertion follows directly from Lemma 4.2 and the definition of $\mathcal{B}$. □

5 The complete ten-integer equation

We now retain the package-31 finalizer. This section gives every formula so that the equation can be read without executing code. The ten unrestricted integer unknowns are exactly

$$\boxed{f, g, h, k, l, w, x, y, z, m}.$$

Neither the 31 source witnesses nor any of the abbreviations below are quantified in the final equation.

5.1 Lucas bridge

Use $b, X, Y, \mathscr{C}$ from Section 4, then define

$$\begin{aligned}
&L = lY, && U = 2LX, && V = 4gwY, && W = bw,\\
&A = U(V+1), && B = 2X+1, && C = B + (A-2)h,\\
&P = U^2V, && K = X + 1 + k(P-2), &&&& (13)\\
&S = 2A - 5, && T = 3WC - 2(W^2-1),\\
&R_2 = (P^2-4)K^2 + 4,
\end{aligned}$$

$$\Delta = 8\mathscr{C}^3 g K^2 - g^2\{32(C-KL)^2\mathscr{C}^3 + g^2K^2\}, \qquad O = f^2 l^2 \Delta.$$

Here the ordinary letter B is different from the radix $\mathcal{B}$. The unknown w is different from the fixed positions w_i. The odd integer $S = 2A - 5$ is always nonzero. Next define

$$\begin{aligned}
&\xi = zx, && D = (A^2-4)C^2 + 4, && E = C^2 D\xi,\\
&F = 4(A^2-4)E^2 + 1, && G^* = 2(A+2)(A-2)^2E^2 - 1, &&(14)\\
&\Omega = 16E, && H^* = C + (B-C)F + \Omega F y,\\
&I^* = ((G^*)^2 - 1)(H^*)^2 + 1.
\end{aligned}$$

These are the formulas of Sun's modified Lucas certificate and binomial bridge [4, Theorems 3.1 and 4.1; Propositions 6.1 and 6.2].

For clarity, the precise imported interface is as follows. The forward bridge, applied to Proposition 4.5, produces f, g, h, k, l, w with b square, R_2 square, $S \mid T$, $C = u_B(A)$, and $O > 0$, where $u_0(A) = 0, u_1(A) = 1, u_{n+1}(A) = Au_n(A) - u_{n-1}(A)$. One may choose f arbitrarily large. On this branch $g \geq b \geq 4$; Δ is a positive multiple of g, so $O \geq 4$. The interpolation theorem permits a positive ξ divisible by a prescribed positive integer and a y in any prescribed residue class modulo ξ, while making DFI^* square.

In the reverse direction, b square, $O > 0$, $\xi \neq 0$, R_2 square, DFI^* square, and $S \mid T$ imply the source conditions of Proposition 4.5(2). The order of the bounds is explicit: $O > 0$ first gives $f, l \neq 0$ and $\Delta > 0$. The square b then gives $b \geq 4$ and $\mathscr{C} > 0$. The displayed formula for Δ forces $K \neq 0$ and $g > 0$; dropping its other nonnegative subtracted term gives $g^3 < 8\mathscr{C}^3$, hence $g < 2\mathscr{C}$. Lemma 4.2 is therefore available before decoding. The Lucas bridge then recovers the required power-of-two and binomial conditions. Inspection of the cited proofs shows that their uses of the source are exactly its bounds, cap shape, forward and reverse implications, and $3 \mid X$. They do not require a recurrence among the coding positions. This is the reason the new adapter composes with them. We invoke these number-theoretic results, rather than claiming to reprove their Lucas-sequence arguments.

5.2 Quintic divisibility and nonvanishing merger

Set

$$\begin{aligned} \pi(v) &= (v^3 - v - 1)(v^2 + 23), & \Sigma &= \xi S^5, \\ r &= T + Sy, & \Theta &= (r^3 - rS^2 - S^3)(r^2 + 23S^2). \end{aligned} \tag{15}$$

Lemma 5.1. *The polynomial π has no rational zero and has a zero modulo every positive integer. If $S \neq 0$ and $\xi \neq 0$, then $\Sigma \mid \Theta$ implies $S \mid T$. If $S \mid T$, some residue class of y modulo ξ makes $\Sigma \mid \Theta$. Moreover $\Theta \neq 0$ for all integer T, y and nonzero S.*

Proof. The cubic is monic and its only possible rational roots are ± 1, neither a root. The quadratic is positive on $\mathbb{Q}$. For an odd prime $p \neq 23$, the cubic has discriminant -23. If it has a root modulo p, that root is simple. Otherwise it is irreducible of degree three, whose Frobenius permutation is an even 3-cycle; its discriminant is then a square modulo p. The quadratic has a simple root. At $p = 23$, the cubic has the simple root 3. Hensel lifting gives roots modulo odd prime powers. Since $-23 \equiv 1 \pmod 8$, it is a square in $\mathbb{Z}_2$, giving roots modulo all powers of 2. The Chinese remainder theorem gives the asserted local roots.

If $S \nmid T$, choose a prime p with $\alpha = v_p(S) > \beta = v_p(T)$. Then $v_p(r) = \beta$. The first homogeneous factor has valuation 3β and the second 2β, because their other terms have strictly larger valuation. Thus $v_p(\Theta) = 5\beta < 5\alpha$, contradicting $S^5 \mid \Theta$.

Conversely write $T = Sq$, choose a root c of π modulo $|\xi|$, and prescribe $y \equiv c - q \pmod \xi$. Then $\Theta = S^5\pi(q + y)$ is divisible by ξS^5. Finally, $\Theta = 0$ would give the rational root r/S of π, which is impossible. □

5.3 One square condition for positivity and the square base

Define

$$\rho_0 = b\{1 + (3O - 4)b^2z^2\}, \qquad \rho_1 = R_2, \qquad \rho_2 = DFI^*. \tag{16}$$

Suppose $z \neq 0$ and ρ_0 is an integer square. The factors b and $1 + (3O - 4)b^2z^2$ are coprime. A nonzero product that is a square forces both to be signed squares of the same sign. But $b \equiv 1 \pmod 3$ cannot be a negative square. The zero-product case likewise forces $|b| = 1$, hence $b = 1$. Therefore b is a positive square and the bracket is a nonnegative square. Since $bz \neq 0$, an integer $O \leq 0$ would make that bracket at most -3. Thus $O > 0$.

Conversely, on the positive bridge branch $O \geq 4$ and b is a square. The positive integer $d = 3O - 4$ is 2 modulo 3, so it is not a square. A positive Pell unit for $u^2 - dv^2 = 1$ can be raised to an exponent making its residue equal to 1 modulo b. This yields a solution with $v > 0$ divisible by b. Write $v = bz$. Then $z > 0$ and ρ_0 is square. Sun's interpolation theorem may now take this z as its prescribed divisibility parameter for ξ; set $x = \xi/z$. Thus the same existing unknown z serves both purposes.

5.4 Separating weights and a radical-free norm

Whenever $z \neq 0$, the actual polynomial ρ_0 satisfies $|\rho_0| > 1$. If $\rho_0 = 0$, then $(3O - 4)b^2z^2 = -1$, forcing $|b| = 1$, hence $b = 1$, $f = 0$, and $O = 0$, a contradiction. If $\rho_0 = \pm 1$, the same argument gives $\rho_0 = 1 - 4z^2 \leq -3$. Also

$$\sqrt{|\rho_1|} \leq P^2 + K^2 + 2,$$

because $|\rho_1| \leq (P^2+4)K^2+4$ and the difference from $(P^2+K^2+2)^2$ is $P^4+K^4+P^2K^2+4P^2 \geq 0$. Set

$$v_0 = 1, \qquad v_1 = \rho_0, \qquad v_2 = \rho_0(P^2 + K^2 + 3). \tag{17}$$

They satisfy

$$|v_1| > \sqrt{|\rho_0|}, \qquad |v_2| > \sqrt{|\rho_0|} + |v_1|\sqrt{|\rho_1|}.$$

In the multiquadratic field generated by the square roots of the ρ_i, a rational signed sum $\sum_i \varepsilon_i v_i \sqrt{\rho_i}$ forces every ρ_i to be an integer square. Indeed, if an automorphism changes a nonzero root term, take the largest changed index. Its magnitude is at least $|v_i|$, larger than the sum of all earlier magnitudes; its change cannot cancel. Hence each root is rational, and a rational square root of an integer is integral. Negative radicands are covered by complex absolute values; zero radicands already are squares.

Finally define

$$\begin{aligned} Z &= \Sigma m - \Theta, & A_i &= \Sigma^2 v_i^2 \rho_i \quad (0 \leq i \leq 2), \\ V_n &= Z^2 + A_2 - A_0 - A_1, & W_n &= V_n^2 + 4Z^2A_2 - 4A_0A_1. \end{aligned} \tag{18}$$

The promised equation is

$$\boxed{F_{10} = W_n^2 - 16V_n^2Z^2A_2 = 0.} \tag{19}$$

Everything in (19) is defined by ordinary polynomial arithmetic above. The proof identity is

$$F_{10} = \prod_{\varepsilon \in \{\pm 1\}^3} \left(Z + \sum_{i=0}^{2} \varepsilon_i \sqrt{A_i} \right).$$

The radicals occur only in this identity, not in the definition of the equation. Expanding the formal norm gives 35 terms.

5.5 Both directions

If the parent positive system has a solution, Proposition 4.5 and the forward bridge give f, g, h, k, l, w, with $O \geq 4$, b, ρ_1 squares, $S \mid T$, and the Lucas identity. The Pell construction gives a positive z making ρ_0 square. Interpolation supplies $\xi > 0$ divisible by z, and a y in the residue class specified by Lemma 5.1, with ρ_2 square. Set $x = \xi/z$. Then $\Sigma \neq 0$ and $\Sigma \mid \Theta$. Choosing

$$m = \Theta/\Sigma - \sum_i \varepsilon_i v_i \sqrt{\rho_i}$$

gives an integer m making one norm factor zero.

Conversely suppose $F_{10} = 0$. Since $S \neq 0$, Lemma 5.1 gives $\Theta \neq 0$. If $\Sigma = 0$, all $A_i = 0$ and (19) becomes $\Theta^8 \neq 0$. Therefore $\Sigma \neq 0$, so $\xi = zx \neq 0$ and $z \neq 0$. A vanishing norm factor, divided by Σ, gives a rational weighted sum of the roots of ρ_i. The separating-weight argument makes all three radicands integer squares. That sum is now integral, so $\Sigma \mid \Theta$. Hence $S \mid T$, while the square ρ_0 gives b square and $O > 0$. The reverse bridge and Proposition 4.5 yield a common nonnegative zero of the shifted residuals. Composing with (2) proves Theorem 1.1.

This proof does not assume that ZFC is consistent, and does not decide whether the displayed equation has a solution. In particular one must not replace the equivalence by an unconditional assertion of no solutions. If ZFC is consistent and the standard hypotheses of the second incompleteness theorem hold, ZFC cannot prove this equation has no integer zero. Consistency alone should not be advertised here as a proof of both directions of independence.

6 Exact degree and circuit size

6.1 A useful degree formula

The source proof applies to m_r quadratic residuals in the same 31 nonnegative coordinates, on a pair-sum-distinct set of maximum M, with the same normalization and coefficient bounds. Then

$$Q = (m_r - 1)(2M + 1) + 4M, \qquad t_N = M + Q + 2 = (2m_r + 3)M + m_r + 1.$$

For the present bridge, direct degree propagation gives

$$\deg F_{10} \leq 6320 t_N + 48M + 1784.$$

For $m_r = 22$, this is

$$297088M + 147144.$$

At $M = 799$, it equals 237 520 456. This explains quantitatively why reducing the source positions matters so much.

For this specific coefficient tuple the relevant leading terms are nonzero. The following table records the main degree propagation, with $t = t_N$:

Polynomial	Degree expression	Value
N	$2t$	75,152
X	$6t$	225,456
Y	$4t$	150,304
$\mathscr{C}$	$2M + 1$	1,599
A	$14t + 3$	526,067
C	$14t + 4$	526,068
P	$24t + 4$	901,828
K	$24t + 5$	901,829
R_2	$96t + 18$	3,607,314
Δ	$56t + 6M + 17$	2,109,067
O	$56t + 6M + 21$	2,109,071
ρ_0	$56t + 6M + 26$	2,109,076
D	$56t + 14$	2,104,270
E	$84t + 24$	3,156,408
F	$196t + 54$	7,364,950
G^*	$210t + 57$	7,891,017
H^*	$280t + 79$	10,521,359
I^*	$980t + 272$	36,824,752

Polynomial	Degree expression	Value
ρ_2	$1232t + 340$	46,293,972
v_2	$104t + 6M + 36$	3,912,734
Σ	$70t + 17$	2,630,337
A_2	$1580t + 12M + 446$	59,380,114
F_{10}	$6320t + 48M + 1784$	237,520,456

In particular A_2 has strictly greater degree than A_0, A_1 and Z^2. The final leading term is A_2^4, with no competitor of its degree. The audit also certifies degree attainment independently by substituting a single indeterminate for all ten unknowns and tracking its leading coefficient modulo three primes:

prime	101	1009	1000003
coefficient of degree 237520456	80	105	684130

A single nonzero entry suffices to prove that the circuit upper bound is attained. This does not require expansion of an astronomically large multivariate polynomial.

6.2 What the operation count charges

The delivered directed acyclic circuit has

222 binary additions, 25 binary subtractions, 562 binary multiplications.

Every multiplication by a literal integer is charged unless removed by a constant identity. Fixed powers are expanded by binary powering; fixed finite geometric sums use

$$G_{2a}(B) = G_a(B)(1 + B^a), \qquad G_{2a+1}(B) = G_{2a}(B) + B^{2a}.$$

No division node occurs. Reuse of a previous node is free. Constant-only arithmetic is folded, as in the predecessor. The 16 fixed recipes are finite integer formulas in the four provided ports; the seventeenth leaf is $2^{2134940}$.

These are arithmetic-circuit counts, not bit-operation counts. They do not include generating the million-bit coefficients, and they say nothing about the size of a hypothetical solution. The circuit is an exact finite specification of an ordinary polynomial, even though printing its full expanded coefficient list would be an unsuitable artifact.

7 Verification, reproducibility, and proof status

7.1 Independent checks supplied

The main verifier reads the delivered circuit, sparse parent, coefficient recipes, and compressed port bytes. It does not import the circuit builder. The retained receipts report:

- all 528 pair sums distinct, including the zero position;
- all 22 shifted quadratics reconstructed symbolically;
- all 136 factorial-weighted coefficient records verified;
- all 22 target-coefficient identities verified as exact polynomial identities in the source variables and fixed ports, including every possible cross-block contribution;
- the four active raw port hashes and coefficient size bound verified by full integer arithmetic;

- 76 named circuit nodes compared against an independent formula transcription at 48 assignments over three finite fields;
- exact degree attained, with the three nonzero leading coefficients displayed above;
- complete small-system source checks, invalid low-digit rejections, a deliberate colliding-position negative control, and arbitrary-g upper-boundary cases;
- regeneration of the native coefficient tuple from the literal transition and tile data, with comparison to the active port hashes.

The symbolic target checks are exhaustive identities. The finite-field node comparisons are transcription tests, not proofs of the unbounded Lucas or compiler statements. The small examples likewise test the source implementation without deciding the existence of an integer zero of the final equation. The main source equivalence is established by the proof in Section 4.

7.2 Reproducing the archive

The top-level `README.md` is the operational entry point. Run

```
python3 reproduce.py
```

to regenerate the circuit and its independent receipts with the Python standard library. Run

```
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

to build the article. The compiler-port reproduction instructions are in `vendor/README.md`. All active data and build inputs are local to the package; no network request or solver query is needed for arithmetic reproduction.

`INSTANCE.json` identifies the authoritative binding and polynomial. `generated/signed10_b2.json` is the complete live-node circuit. `generated/source_records.json` contains the coefficients needed to interpret every fixed leaf. `MANIFEST.sha256` records the delivered files. Proof notes retain the independent source reviews, while `vendor/` records the inherited material and attribution.

The repository baseline is pinned at `22e75dd6b87b1b199bb08c778d7cf075ae5aae08`. Some inherited files name earlier commits for their own sources; these are provenance records, not instructions to mix coefficient tuples. In particular any default binding metadata in the raw parent sparse file is superseded by the explicit whole-tuple binding in this package.

7.3 What remains outside the certificate

The new source adapter and its algebraic instantiation are given in full. The semantic machine/compiler chain and Sun's 2026 Lucas/interpolation theorems remain named mathematical dependencies. The latter source is a 2026 preprint; this report does not supply a new formal verification of it. The archive includes the endpoint reference and inherited proof material, but not a rebuild of the entire ProveIt proof-assistant project. Thus the correct status is an ordinary proof with explicit dependencies and extensive finite algebraic verification.

No conclusion here establishes the truth or falsity of Con(ZFC). Nor do finite tests establish a new independence theorem. They certify the explicitly stated arithmetic construction and its transcription.

8 Further reductions suggested by the construction

The practical change is to optimize the source compiler before optimizing the final norm. It suggests several concrete directions.

Support-aware positions. A full B_2 set separates every quadratic monomial, whereas the actual 22 quadratics use only a small subset. It may suffice to exclude collisions between a source-supported monomial and any multiplier monomial that could reach its target. A smaller list could lower M, and the present degree formula would lower the degree by 297088 for each unit saved. Such a construction needs an explicit collision certificate; simply dropping unused pairs without auditing all multiplier monomials is unsound.

Support-aware targets. Uniform stride $2M + 1$ is a simple, proved sufficient condition. The individual reversed blocks often have sparse or shorter support. Nonuniform target positions could reduce their total height, provided every cross-block contribution to a marked coefficient is excluded. The current independent symbolic extraction checker is a suitable certificate format for that search.

Better small source systems. Reducing the number of source variables changes the size of the required additive set; reducing the number of residuals shortens the block range. These objectives may trade against source degree, coefficient size, and evaluation cost. A smaller positive system cannot be substituted without its complete sign, normalization, and compiler-interface proof.

Higher-degree simultaneous coding. The coefficient-extraction argument generalizes to a B_d set, with the zero position representing padding to degree d. Weights $(d - |\alpha|)! \prod_i \alpha_i!$ extract $d!\, r_j$; stride $dM + 1$ isolates targets. This is an algebraic generalization of Lemma 4.1. A complete higher-degree source theorem must separately redo the coefficient, carry, and reverse-branch bounds; it should not be asserted solely from the algebraic identity.

Formalization. The cleanest next formal target is the source interface: finite monomial separation, the target-coefficient identity, binary masks, and the binomial valuation lemma. These are independent of K215 and can be stated as reusable arithmetic lemmas. The existing circuit and record format then permits a small certificate checker. Formalizing the complete result also requires the inherited Lucas and compiler dependencies.

Nine variables and global optimization. The result does not give a nine-unrestricted-integer equation. Nor does it automatically improve the repository's nine-positive-variable construction: that uses a different compressor whose interface must be re-established. Such extensions are research tasks, not consequences of a variable-name substitution.

A The pinned positive source system

The 31 strictly positive variables, in the order used by the coefficient records and weights $w_1, \dots, w_{31}$, are

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,\\ N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta).$$

These names belong to the *parent* system and are local to this appendix; they are not identified with the final ten unknowns. Define the affine abbreviations

$$q = \beta_{\mathrm{native}} J + 1, \qquad k = \eta + \zeta, \qquad H = 4a + 3, \qquad C = q - F - \alpha.$$

The 22 equations are

$$\begin{aligned}
L &= q^2 - 1,\\
N &= q(L + 1),\\
X &= wq,\\
a &= E + sN,\\
E &= X(a - E),\\
c &= k(a - E) + \eta,\\
P &= E(c - \eta) + k,\\
\tau^2 &= P(P - k) + 1,\\
D &= X + ac + \gamma H,\\
\Delta &= a^2 + H,\\
C_2 &= c^2,\\
D^2 &= \Delta C_2 + 1,\\
k &= jc - V + 1 + hE,\\
jc - V &= (q - F)(N - q) - CL + (M_C + qM_F)J,\\
(K_0 + w)C + (q - F) - z(q - 1) &= 1,\\
F_2 &= f^2,\\
V &= of - c,\\
g &= y^2 - V^2,\\
I &= iC_2,\\
B_{\mathrm{aux}} &= I^2,\\
B_{\mathrm{aux}} &= \Delta(F_2 - 1),\\
y^2 - B_{\mathrm{aux}} g &= 1.
\end{aligned}$$

Residuals mean left side minus right side after the affine abbreviations are expanded. They are quadratic in these 31 variables, with the four compiler ports treated as constants. The copied sparse file has SHA-256

`1c336ba0b3c9827ffddcf7ce895cad9dca2907f907cde7484c7f387bb2cb1287.`

B Coefficient binding and exact bytes

The active compiler has

$$\rho = 2^{121}, \qquad L = 8801, \qquad d = 121 \cdot 8801 = 1064921, \qquad B_{\text{native}} = 2^d.$$

Its entire finite recipe, transition table, tile table, and controller proof are included in `vendor/`. The unsigned integers are encoded little-endian and gzip-compressed in `data/`. The raw integer hashes are:

β_{native}**: `beta.bin.gz`, 1,064,921 bits**
`6a029139964239dc65d716ccc186b40d9c5923ada0a4116cbb418d84534096ea`

K_0**: `K.bin.gz`, 1,070,004 bits**
`6984850dcfc65498b42b633855ea81b8c2cf6d0a84e49005cca10f25557ae395`

M_C**: `MC.bin.gz`, 1,064,921 bits**
`7c2b34179cb98c9bea1954f816f93fdc1f60213c68d8e98f1541edb41f0e3a4b`

M_F**: `MF_source.bin.gz`, 1,064,922 bits**
`28c55970993394dd7ee32d7299a215fd26b4d70e437a643b60594eee8c7cd3a7`

The active source constant is $\beta_S = 2^{2134940}$. It is distinct from $\beta_{\text{native}} = B_{\text{native}} - 1$. The final equation's fixed leaves are specified completely by these four integers, that explicit power of two, the finite coefficient recipes, and literal small integers.

C A proof dependency map

Step	Statement used	Status in this package
K215 endpoint	Blank halting iff a finite ZFC refutation	Inherited endpoint source and precise theorem name
Native compiler and positive arithmetic	The 31-positive-variable system has a solution iff K215 halts	Inherited ordinary proofs, literal inputs, regenerated coefficient tuple
Direct residual source	Common quadratic zero iff the binary/binomial source interface	New proof, full symbolic coefficient checks, exact data
Lucas and interpolation bridge	Source interface iff the specified square/divisibility certificate	Sun 2026 theorems, same formulas as package 31
Final mergers and norm	Certificate iff one equation in ten integers	Inherited package-31 arguments reproduced in this report
Degree and circuit transcription	Exact degree 237520456, 809 operations	Independent arithmetic audit and leading-coefficient certificate

References

[1] V. Reshetnikov and contributors, *ProveIt: Con(ZFC) Diophantine forms collection*, audited at commit `22e75dd6b87b1b199bb08c778d7cf075ae5aae08`, 23 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/22e75dd6b87b1b199bb08c778d7cf075ae5aae08/SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/con-zfc-diophantine-forms.

[2] ProveIt, package 31, ten-integer source, coefficient compiler, and structural-weight quintic finalizer. Pinned source copies and notices are included in this archive. See the collection in [1].

[3] Zhi-Wei Sun, *Further results on Hilbert's Tenth Problem*, Science China Mathematics **64** (2021), 281–306. Version consulted: arXiv:1704.03504v7. https://arxiv.org/html/1704.03504v7.

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[5] R. C. Bose and S. Chowla, *Theorems in the additive theory of numbers*, Commentarii Mathematici Helvetici **37** (1962/63), 141–147. https://doi.org/10.1007/BF02566968.

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